

Problem 1.

Solution: From the question, we are given the infeasible LP:

$$\max c^T x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0.$$

The we can augment the RHS by purchasing $\lambda_i \in [0, k_i]$ for constraint i at unit cost d_i . Let $\lambda \in \mathbb{R}_{\geq 0}^m$, $d, k \in \mathbb{R}_{\geq 0}^m$. The augmented constraints are $Ax \leq b + \lambda$ and the augmentation cost is $d^T \lambda$. We then just have to solve for:

$$\max_{x, \lambda} c^T x - d^T \lambda : Ax \leq b + \lambda, \quad (1)$$

$$0 \leq \lambda \leq k, \quad (2)$$

$$x \geq 0. \quad (3)$$

Through solving this single LP, we can simulate the process of buying λ , with price $d^T \lambda$. Then solving the augmented LP gives the maximum of the net value $c^T x - d^T \lambda$.

Problem 2.

Solution: From the question we can find the feasible region:

$$P = \{x : Ax = b, x \geq 0\} \quad (4)$$

primary: solve the optimization for

$$\max \{ c^T x : Ax = b, x \geq 0 \} \quad (5)$$

Let the optimal value be $z^* = \max c^T x : x \in P$.

secondary: solve the optimization for

$$\max \{ d^T x : Ax = b, x \geq 0, c^T x = z^* \}. \quad (6)$$

Returning any optimizer for (6) is could be the desired solution. The feasible set of (6) is exactly the set of all solutions that maximize $c^T x$ over P .

Problem 3.

Solution:

1) Complete network, no Capacities

Necessity: If $x = (x_{uv})_{(u,v) \in A}$ is a feasible flow, then each node v satisfies the flow balance

$$\sum_{u: (u,v) \in A} x_{uv} - \sum_{w: (v,w) \in A} x_{vw} = d_v.$$

Summing these equalities over all $v \in V$ cancels every arc variable once with $+1$ (as inflow) and once with -1 (as outflow), yielding

$$\sum_{v \in V} d_v = 0.$$

Therefore $\sum_v d_v = 0$ is necessary for feasibility.

Sufficiency: We can assume $\sum_{v \in V} d_v = 0$. Because the digraph is complete and arcs have infinite capacity, send flow directly from supplies to demands by the following pairing procedure. Let $S = \{i \in V : d_i > 0\}$ and $T = \{j \in V : d_j < 0\}$. Initialize all flows to 0. While S and T are nonempty, pick $i \in S$, $j \in T$, send $f = \min\{d_i, -d_j\}$ units on arc (i, j) , and update $d_i \leftarrow d_i - f$ and $d_j \leftarrow d_j + f$. Each iteration zeroes at least one of d_i or d_j , so the process terminates in finitely many steps. At termination all modified d 's are zero, so the constructed x satisfies every node balance; capacities are respected (they are infinite). Thus a feasible flow exists.

Therefore, on a complete network without capacities, $\sum_{v \in V} d_v = 0$ is both necessary and sufficient for feasibility.

2) Capacity

The condition $\sum_{v \in V} d_v = 0$ remains necessary but is not sufficient. For example, if we take two nodes u, v with $d_u = 2$, $d_v = -2$, arcs in both directions with capacities $u \rightarrow v$ of 1 and $v \rightarrow u$ of 1. Here $\sum d_v = 0$, but no feasible flow can deliver the 2 units from u to v because the cut capacity out of $\{u\}$ is only 1.

In general, with capacities, feasibility is characterized by the cut conditions for every subset $S \subseteq V$, the net supply across the cut cannot exceed the forward cut capacity, $\sum_{v \in S} d_v \leq u(\delta^+(S))$, and similarly for the reverse cut.

Problem 4.

Solution:

- a) LP formulation Let $p_t \geq 0$ be production in period t and $s_t \geq 0$ be end-of-period inventory.

$$\begin{aligned} \min \quad & \sum_{t=1}^T c_t p_t + \sum_{t=1}^{T-1} h_t s_t \\ \text{s.t.} \quad & s_{t-1} + p_t = d_t + s_t \quad (t = 1, \dots, T), \quad s_0 = 0, \quad s_T = 0, \\ & 0 \leq p_t \leq u_t \quad (t = 1, \dots, T), \quad 0 \leq s_t \leq q_t \quad (t = 1, \dots, T-1). \end{aligned}$$

The inventory after T has no value and only adds cost, so an optimal solution has $s_T = 0$.

- b) Based on the randomly generated data points, the data points:

- *Production tracks demand.* With positive holding costs, the model prefers to produce in the same period as the demand whenever capacity allows.
- *Inventory is minimal.* $s_t = 0$ in most periods; it becomes positive mainly when future demand exceeds available production capacity, so the model builds stock earlier.

- *Inventory is used strategically.* When capacity is binding, inventory acts as a buffer to shift production from lower-cost or slack periods to cover later demand.
- *Final-period condition.* $s_T = 0$ in optimum since holding beyond the horizon yields no benefit.

Problem 5.

Solution:

a) We are given:

$$\begin{aligned} \max \quad & 2x_1 - x_2 + x_3 \\ \text{s.t.} \quad & 3x_1 + x_2 + x_3 \geq 40 \end{aligned} \tag{1}$$

$$x_1 - x_2 + 2x_3 = 15 \tag{2}$$

$$x_1 + x_2 - x_3 \leq 30 \tag{3}$$

$$x_1 \geq 0, \quad x_2 \leq 0,$$

Dual:

Let y_1 correspond to (1), y_2 to (2), y_3 to (3). Because the primal is a maximization problem, column sign rules give:

$$\begin{aligned} \min \quad & 40y_1 + 15y_2 + 30y_3 \\ \text{s.t.} \quad & 3y_1 + y_2 + y_3 \geq 2 \quad (x_1 \geq 0) \\ & y_1 - y_2 + y_3 \leq -1 \quad (x_2 \leq 0) \\ & y_1 + 2y_2 - y_3 = 1 \quad (x_3 \text{ free}) \\ & y_1 \leq 0, \quad y_3 \geq 0, \end{aligned}$$

b) **Solving the dual:**

From the equality: $y_1 = 1 - 2y_2 + y_3$. Substituting into the two inequalities yields

$$5y_2 - 4y_3 \leq 1, \quad 3y_2 - 2y_3 \geq 2,$$

with $y_3 \geq 0$ and $y_1 \leq 0 \iff y_3 \leq 2y_2 - 1$. The optimum occurs where both are tight: solving $5y_2 - 4y_3 = 1$ and $3y_2 - 2y_3 = 2$ gives

$$y_2^* = 3, \quad y_3^* = \frac{7}{2}, \quad y_1^* = 1 - 2 \cdot 3 + \frac{7}{2} = -\frac{3}{2}.$$

Dual optimum value:

$$40y_1^* + 15y_2^* + 30y_3^* = 40\left(-\frac{3}{2}\right) + 15 \cdot 3 + 30 \cdot \frac{7}{2} = 90.$$

Primal optimum. At optimality the three primal constraints are tight (because $y_1^* \neq 0$, $y_3^* \neq 0$, and (2) is an equality). Solving

$$\begin{cases} 3x_1 + x_2 + x_3 = 40, \\ x_1 - x_2 + 2x_3 = 15, \\ x_1 + x_2 - x_3 = 30, \end{cases} \Rightarrow x_1^* = 40, \quad x_2^* = -45, \quad x_3^* = -35.$$

Signs are satisfied ($x_1^* \geq 0$, $x_2^* \leq 0$, x_3^* free) and

$$2x_1^* - x_2^* + x_3^* = 2 \cdot 40 - (-45) + (-35) = 90.$$

Both primal and dual are feasible and bounded, with optimal at value 90, which shows strong duality

Problem 6.

Solution:

- a) **Primal:** Denver D , Chicago C , Houston H , New York N , Miami M . Arcs and data (first number = capacity u , second = unit cost c):

$$\begin{aligned} D \rightarrow C : (u_{DC}, c_{DC}) &= (3, 1), & D \rightarrow H : (2, 6), \\ C \rightarrow H : (4, 2), & C \rightarrow N : (2, 3), & H \rightarrow M : (1, 4), & N \rightarrow M : (2, 5). \end{aligned}$$

Node supplies/demands d_v (positive = demand, negative = supply):

$$(d_D, d_C, d_H, d_N, d_M) = (-4, -2, 3, 1, 2), \quad \sum_v d_v = 0.$$

Variables $x_a \geq 0$ for each arc a :

$$\begin{aligned} \min & c_{DC}x_{DC} + c_{DH}x_{DH} + c_{CH}x_{CH} + c_{CN}x_{CN} + c_{HM}x_{HM} + c_{NM}x_{NM} \\ \text{s.t.} & \begin{cases} -x_{DC} - x_{DH} = d_D = -4, \\ x_{DC} - x_{CH} - x_{CN} = d_C = -2, \\ x_{DH} + x_{CH} - x_{HM} = d_H = 3, \\ x_{CN} - x_{NM} = d_N = 1, \\ x_{HM} + x_{NM} = d_M = 2, \end{cases} \\ & 0 \leq x_{DC} \leq 3, \ 0 \leq x_{DH} \leq 2, \ 0 \leq x_{CH} \leq 4, \ 0 \leq x_{CN} \leq 2, \\ & 0 \leq x_{HM} \leq 1, \ 0 \leq x_{NM} \leq 2. \end{aligned}$$

Dual:

Let node potentials π_v (free) correspond to the five flow-balance equalities, and let $\mu_a \geq 0$ correspond to the six upper-bound constraints $x_a \leq u_a$. The dual is

$$\begin{aligned} \max & -4\pi_D - 2\pi_C + 3\pi_H + \pi_N + 2\pi_M - 3\mu_{DC} - 2\mu_{DH} - 4\mu_{CH} - 2\mu_{CN} - 1\mu_{HM} - 2\mu_{NM} \\ \text{s.t.} & \pi_C - \pi_D - \mu_{DC} \leq 1, \\ & \pi_H - \pi_D - \mu_{DH} \leq 6, \\ & \pi_H - \pi_C - \mu_{CH} \leq 2, \\ & \pi_N - \pi_C - \mu_{CN} \leq 3, \\ & \pi_M - \pi_H - \mu_{HM} \leq 4, \\ & \pi_M - \pi_N - \mu_{NM} \leq 5, \\ & \mu. \geq 0, \quad \pi. \text{ free.} \end{aligned}$$

Each constraint corresponds to an arc (i, j) : $\pi_j - \pi_i - \mu_{ij} \leq c_{ij}$.

- b) **Generic min-cost flow:** Given a directed network $N = (V, A)$, costs c_a , capacities $u_a \geq 0$, and node supplies/demands d_v with $\sum_{v \in V} d_v = 0$, the standard LP is

$$\begin{aligned} \min \quad & \sum_{a \in A} c_a x_a \\ \text{s.t.} \quad & \sum_{a=(u,v) \in A} x_a - \sum_{a=(v,w) \in A} x_a = d_v \quad \forall v \in V, \\ & 0 \leq x_a \leq u_a \quad \forall a \in A. \end{aligned}$$

Generic dual:

The node potentials π_v free; capacity multipliers $\mu_a \geq 0$ is

$$\begin{aligned} \max \quad & \sum_{v \in V} d_v \pi_v - \sum_{a \in A} u_a \mu_a \\ \text{s.t.} \quad & \pi_{\text{head}(a)} - \pi_{\text{tail}(a)} - \mu_a \leq c_a \quad \forall a \in A, \\ & \mu_a \geq 0 \quad \forall a \in A, \quad \pi_v \text{ free } \forall v \in V. \end{aligned}$$

Equivalently, the reduced-cost condition is $\pi_j - \pi_i \leq c_{ij}$ when the arc is not tight ($\mu_{ij} = 0$), and is relaxed by μ_{ij} when the upper bound binds.